

Hybrid-Electric Design Studies for a Long-Endurance Tailsitter Concept

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A conceptual-level study was conducted to compare suitability of four different propulsion systems for long endurance tailsitter aircraft. Diesel, parallel hybrid diesel, turboshaft, as well as advanced series hybrid solid oxide fuel cell designs were considered. Each configuration was optimized and loitering time varied to quantify performance and high-level design characteristics as the mission changed. Design excursions showing gross weight sensitivity to varying range were also performed. The hybrid-electric configurations allowed for greater design flexibility by partially decoupling hover power requirements from energy conversion system size, resulting in substantially differing optimal airframes and forward flight speeds compared to the single engine aircraft. Results were dependent on best-in-class electric motor technology, batteries concurrently possessing high C-rate and high pack-level specific energy, and a duty cycle consisting of a short hover period and sustained operation at part power. A key insight from this study is that, with the appropriate battery and electric motor technology, hybrid-electric propulsion architectures can enable VTOL capable, long-range, and fuel-efficient aircraft for low-speed applications by pairing fuel-efficient engine choices with power dense electric motors and batteries. Solid oxide fuel cells in particular exhibit promising characteristics for these use cases, but widespread adoption in aviation applications requires the maturation of emerging higher specific power technology or combined cycle approaches.

I. Nomenclature

CW/s	=	rotor blade loading
DGW	=	design gross weight (lb)
DL	=	disk loading (psf)
Esp	=	specific energy (MJ/kg)

Distribution Statement A: Approved for public release; distribution unlimited. PR2024357

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Presented at the 2025 AIAA Scitech Forum, Orlando, FL, January 6-10, 2025. AIAA 2025-1436

$E_{sp_{eff}}$	=	effective specific energy (MJ/kg)
MCP	=	maximum continuous power
MRP	=	maximum rated power
P_{sp}	=	specific power (kW/kg)
$P_{sp_{bop}}$	=	balance of plant specific power (kW/kg)
$P_{sp_{fuelcell}}$	=	fuel cell stack specific power (kW/kg)
SFC	=	brake specific fuel consumption (lb/hp-hr)
SLS	=	sea level standard conditions
SOFC	=	solid oxide fuel cell
WL	=	wing loading (psf)
W_{BOP}	=	balance of plant weight (lb)
W_{FC}	=	fuel cell weight (lb)
η_{energy}	=	energy conversion efficiency
η_{tank}	=	tank efficiency

II. Introduction

There is continued investment in a wide range of UAS aircraft types within the Army with an interest in assessing the impact of the propulsion architecture on performance, weight, cost, and military suitability. Hybrid-electric configurations have generated considerable interest, in part a result of the success of hybrid-electric automobiles. Reasons for potential hybrid propulsion system benefits are very different for aircraft than those for automobiles. Automotive hybrid systems typically incorporate regenerative braking which allows the vehicle to recover energy from the frequent stop/start cycles that would be lost in a more conventional friction brake internal combustion engine system by using a battery. An analogous phenomenon in aircraft involves potential for windmilling to allow the rotor/propeller to convert airflow into electrical energy and store in a battery. Potential windmilling benefits for the vast majority of aircraft are considerably smaller than for regenerative braking for automobiles as aircraft do not typically see the frequent stop-start duty cycles common in automobiles that operate in more urban environments. Another important impact is engine design/selection; Atkinson Cycle-based engines are commonly selected for hybrid-electric automobiles, which provide higher efficiency at the expense of lower power density compared to the more commonly utilized Otto Cycle-based engines.

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In aircraft, particularly, rotorcraft, hybrid-electric drivetrains may present opportunities to tailor the propulsion system to meet mission requirements, which, in some instances, may result in lower overall systems weight. Power requirements in hover are sometimes 2-3X greater than that in cruise or loiter, yet typical hover durations are brief (~30 seconds to 5 minutes) [1]. As a result, many rotorcraft carry a propulsion system that operates at part power through most of its operational lifetime. This may present opportunities to reduce the size and tailor an engine for higher efficiency at lower power levels with an additional power system for short-duration, high power flight states. Figure 1 illustrates a qualitative power bucket (representative for VTOL designs) to illustrate this opportunity.

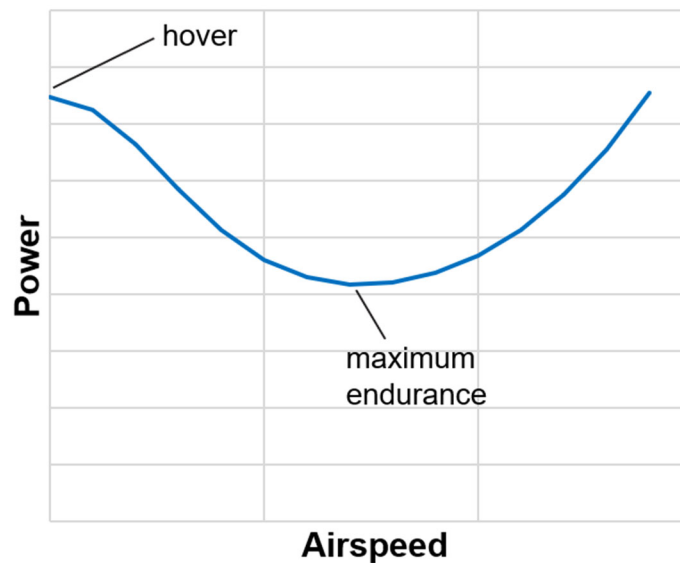


Fig 1: Generic Power-Speed Profile

The analogous approach in a parallel hybrid automotive system is that the system, to a first order approximation, runs the engine at an efficient design point to keep the battery charged while employing the battery and electric motor during high power transient operation [2]. In reality, operation of the hybrid-electric system involves complex system level optimization based on battery state of charge, time integrated engine loading, etc.; performing similar aircraft-level analysis at the conceptual design level is not appropriate. Most aircraft are unlikely to see the frequent mid-mission start-stop cycles common in automotive operation, but some may benefit from further opportunities to integrate a heavier, more fuel-efficient system depending on the set of mission requirements.

Recent events have spurred the Department of Defense to pivot and direct a larger share of available funding to unmanned concepts [3]. For unmanned applications, tailsitters present opportunities to leverage the comparative

forward flight efficiency of fixed wing aircraft while maintaining the runway independence and operational flexibility of vertical takeoff and landing aircraft [4]. In addition, they offer reduced mechanical complexity for conversion compared to tiltrotor and tiltwing aircraft. This paper investigates potential design characteristics for hybrid-electric drivetrains applied to an unmanned long-endurance coaxial tailsitter concept. A prior study optimizing aircraft for a reference military mission generated turboshaft-powered designs with lower fuel burn and DGW than diesel and hybrid-electric diesel options due to the superior power/weight characteristics of gas turbine engines [5]. For this paper, a long-endurance mission was constructed to explore applications for which hybrid arrangements are more likely to provide useful capability via the aforementioned propulsion system tailoring.

III. Approach

VTOL loitering aircraft (such as those explored in this paper) spend most of their design mission at partial power, which for conventional turboshaft engines results in higher specific fuel consumption (SFC) and therefore lower energy conversion efficiency. Tailsitter configurations were selected for this study because of their potential for both high forward flight efficiency and VTOL capability, presenting design flexibility for different propulsion system options. Additionally, the propulsion actuation system was mechanically actuated, allowing for consistent comparisons when utilizing these propulsion system choices. A simple OpenVSP representation of a baseline turboshaft-powered aircraft is shown in Fig 2 [6]. This baseline aircraft possessed a wing loading of 15 psf, aspect ratio of 10, rotor disk loading of 6 psf, and rotor tip speed of 750 ft/s. Weight and aerodynamic assumptions were derived from Ref. [5]. Wing loading and aspect ratio were modified from Ref. [5] to improve suitability for long endurance, low-speed missions. Vehicle analyses were performed using NDARC [7]. Wing weight was calculated using the AFDD93 regression, which increases with wing area and structural design gross weight. Rotor weight was estimated using the AFDD82 regressions based on rotor radius, chord, and tip speed. Fuselage weight was estimated using the AFDD84 universal body weight equation, which is a function of aircraft weight. Wing weight, rotor weight, and fuselage each varied with other parameters which remained constant across the aircraft analyzed in this study.

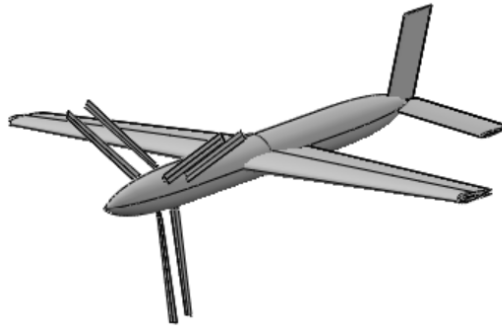


Fig 2: Baseline Aircraft

All aircraft in this study possessed the same fuselage and empennage geometry. Empennage geometry was selected for ground stability prior to takeoff; suitability for takeoff/landing in windy conditions was not investigated. Empennage weight was estimated using the AFDD82 helicopter parametrics which are functions of tail area and aspect ratio but not design gross weight. As a result, estimated empennage weight was identical for each of the aircraft in this study. Geometry requirements for the fuel tank were not considered in this paper but are a planned topic of future work. Table 1 shows the vehicle geometric parameters that remained constant for all aircraft analysis.

Table 1: Common Vehicle Geometry

Parameter	Value
Fuselage	
-wetted area (ft ²)	350
-length (ft)	33
-maximum height (ft)	3.5
-maximum width (ft)	4.3
Wing	
-taper ratio	0.5
-t/c	0.18
-aspect ratio	10
Horizontal Tail	
-area (ft ²)	58
-aspect ratio	5.6
Vertical Tail	
-area (ft ²)	27
-aspect ratio	2.37

Four propulsion configurations were considered: a pure diesel propulsion system, a diesel parallel hybrid propulsion system, a pure turboshaft propulsion system, and a series-hybrid solid oxide fuel cell (SOFC) propulsion

system. A diagram depicting each propulsion system arrangement can be seen in Fig. 3. Throughout this paper, “diesel-electric” is used as shorthand to refer to the hybrid-electric diesel architecture.

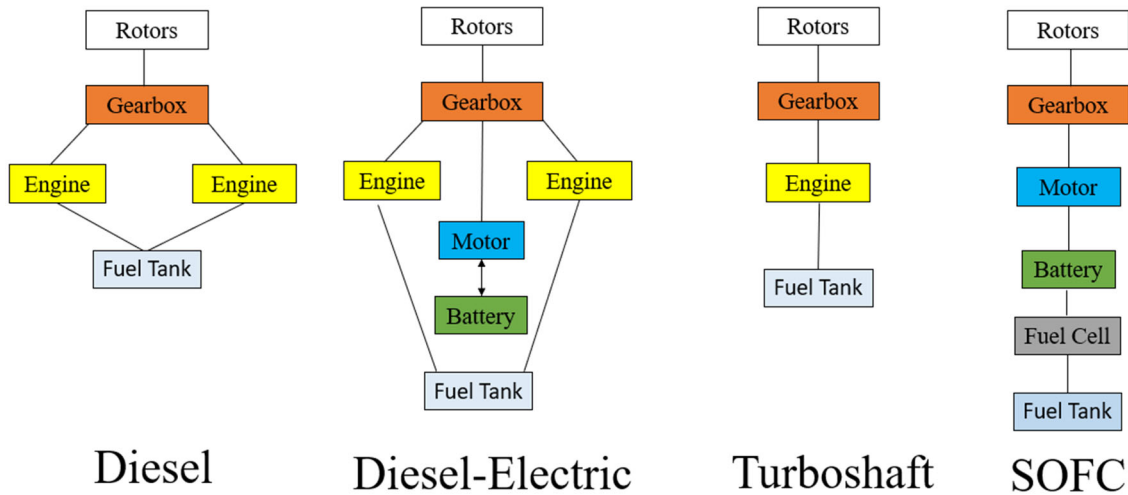


Fig. 3: Propulsion System Arrangements

The diesel and diesel-electric arrangements were turbocharged, with two diesel engines arranged in parallel such that the engine power remains close to that of currently available systems. Diesel engine performance in this paper assumed a power/weight ratio and a reference SFC that did not change with engine size. Electric motor scaling was based on a regression from Ref. [8] calibrated to published peak specific torque performance from Ref. [9] (20 ft-lb/lb) at a 1,000 hp reference condition. Peak torque in motors may be as high as 2-3X the continuous rating, but operation at this condition may be limited to ~30 seconds due to thermal limitations. As a result, all electric motors in this paper were sized to handle continuous operation for all mission conditions. For this paper, continuous torque was assumed to be 45% of peak torque. Motor efficiency was fixed at 95%. Motors are expected to operate near peak efficiency across a wide range of conditions, rapidly declining at very low rotational speeds and/or torques [10]. For the diesel-electric architecture, where the motor is only operational during short hover operation, this is a very good assumption for a well-selected motor system. The motor in the SOFC architecture operated under a wider range of conditions, but well within the typical higher-efficiency motor region. Power electronics weight and performance assumed high voltage DC architectures similar to those envisioned for emerging all-electric aircraft for Urban Air Mobility applications [11]. Transmission losses were assumed as a constant 4%. A single engine was used for the turboshaft designs to take advantage of beneficial scaling effects. Turboshaft engine scaling for weight

and SLS SFC were based on the below equations (from Ref. [12]). Maximum rated power (P_{MRP}) is in hp, W_{dry} is in lb and SFC is in lb/hp-hr.

$$W_{dry} = 4E - 6 * P_{MRP}^2 + 0.06 * P_{MRP} + 190 \quad (1)$$

$$SFC = 0.9 * (P_{MRP}^{-0.11}) \quad (2)$$

SOFCs are electrochemical devices that convert fuel into electrical energy. SOFCs possess low specific power as well as long startup times. To ensure transient maneuver response capability, all SOFCs were in a series hybrid arrangement, providing electrical power to a battery that in turn provides electrical power to an electric motor. Solid oxide fuel cells possess a variety of potential feedstock options, including hydrogen, heavy fuels, methane, and propane. For heavy fuels such as diesel and JP8, this fuel flexibility is dependent on external reformation as well as desulfurization for extended operation without performance loss. Propane has been demonstrated as a fuel for SOFC-powered aircraft on multiple systems and may be reformed internally, eliminating much of the heavy ancillary equipment [11, 12, 13]. Different arrangements utilizing combinations of compressors, turbines, combustors, and SOFCs have been proposed [14, 15]. For this paper, specific power of the fuel cell balance of plant was estimated via the following:

$$W = W_{fuelcell} + W_{BOP} \quad (3)$$

$$\frac{W}{P} = \frac{W_{fuelcell} + W_{BOP}}{P} = \frac{1}{Psp} \quad (4)$$

$$\frac{1}{Psp} = \frac{1}{Psp_{fuelcell}} + \frac{1}{Psp_{BOP}} \quad (5)$$

$$Psp = \frac{1}{\frac{1}{Psp_{fuelcell}} + \frac{1}{Psp_{BOP}}} \quad (6)$$

For current trends, approximately 75% of the balance of plant (BOP) weight is estimated to come from the fuel cell stack [16]. A combined 1.2 kW/kg for the balance of plant was used based on demonstrated systems and includes a compressor, turbine, cooling pumps, and other ancillary equipment [18]. The specific power of the fuel cell stack was 2.5 kW/kg based on projected numbers from a NASA patent for a bi-electrode supported SOFC [17-19]. This yields a net system-level specific power of 0.81 kW/kg. Results for this paper assume an advanced, high specific power SOFC system but appear to be achievable (or surpassable) with a combined cycle approach [16]. For

completeness, a brief description of a notional combined cycle approach is included. Combined cycle approaches offer synergistic benefits beyond balancing power, efficiency, and weight as the simplified diagram in Fig. 4 illustrates.

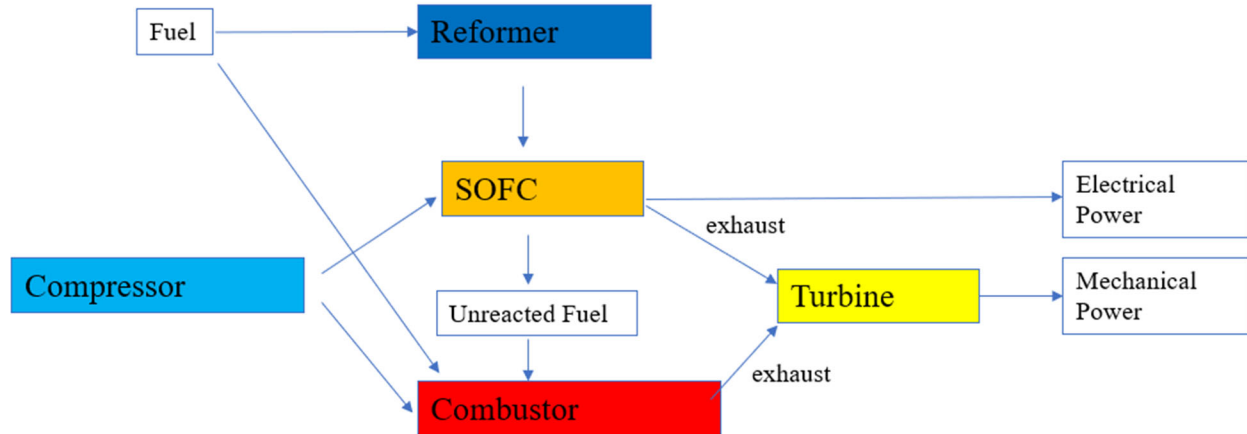


Fig. 4: Simplified Combined Cycle Schematic

SOFCs operate at high temperatures, ranging from 600-800°C; recovering and supplying this hot, pressurized exhaust to a turbine extracts additional energy from the fuel and increases system-level efficiency [22]. In addition, practical fuel utilization in SOFCs ranges from 70-85% [23]. Burning unutilized fuel in a combustor and venting to power a gas turbine further increases the system-level power output and energy conversion efficiency in exchange for a comparably small increase in weight. Fig. 4 is a simplification of the system to illustrate potential synergies; in reality, these proposed combined cycle approaches usually include recuperation to further increase system efficiency as in Refs. [22-23]. Challenges for combined cycle systems include complexity and restricted operating range of the system due to SOFC-gas turbine component-level coupling [24, 25].

Combined cycle approaches, to the best of the author's knowledge, have not been demonstrated on aircraft at the time of writing. However, numerous system design studies for aviation applications exist [14, 15, 20, 25 -27]. Moreover, commercially available ground power systems have shown total combined heat and power energy conversion efficiencies greater than 90% [30]. To align with currently fielded systems, all SOFC-powered aircraft in this paper utilize propane, which reduces reformation and desulfurization requirements. A summary of representative performance parameters for the energy systems explored in this hybridization study are shown in

Table 2. The diesel, SOFC, and battery properties were fixed for each aircraft. Energy tank efficiency and effective specific energy are represented by η_{tank} and Esp_{eff} . They are defined as:

$$\eta_{\text{tank}} = \frac{W_{\text{fuel}}}{W_{\text{fuel}} + W_{\text{tank}}} \quad (7)$$

$$\text{Esp}_{\text{eff}} = \text{Esp} * \eta_{\text{tank}} * \eta_{\text{energy}} \quad (8)$$

Table 2: Energy System Comparison

Propulsion System	Psp (kW/kg)	Esp (MJ/kg)	η_{tank}	η_{energy}	Esp_{eff} (MJ/kg)
Turboshaft (800 hp)	4.60	42.6	0.90	0.25	9.59
Diesel	0.85	45.6	0.90	0.37	15.18
SOFC (Propane)	0.81	50.3	0.75	0.60	22.64
Battery	4.50	1.08	0.75	0.9	0.73

Relevant metrics for comparison to inform potential hybrid-electric architecture candidates are specific power (Psp) and effective specific energy (Esp_{eff}). Esp_{eff} accounts for both energy conversion efficiency as well as tank efficiency; the latter becomes more relevant when comparing fuels that require pressurization (such as propane) as opposed to liquid heavy fuels. Turboshaft engines possess higher specific power than the other energy systems in Table 2 and convert chemical energy into mechanical energy which is transmitted to the rotor via a gearbox. Diesel engines at these scales possess higher conversion efficiency, but lower power/weight than the turboshaft and also transmit mechanical energy to the rotor via a gearbox. SOFCs are significantly more efficient than either diesel or turboshaft engines. However, current systems possess lower power-to-weight characteristics than the other propulsion system options in this paper. In addition, because SOFCs output electrical, rather than mechanical power, they require an electric motor to provide mechanical power for propulsive thrust as well as a battery to ensure suitable operation in transient flight states. In Table 2, η_{tank} for the battery refers to the battery management system. Esp_{eff} and Psp for the energy systems in Table 2 are depicted graphically in Fig. 5 to illustrate potential energy system combinations relevant for this hybridization use case.

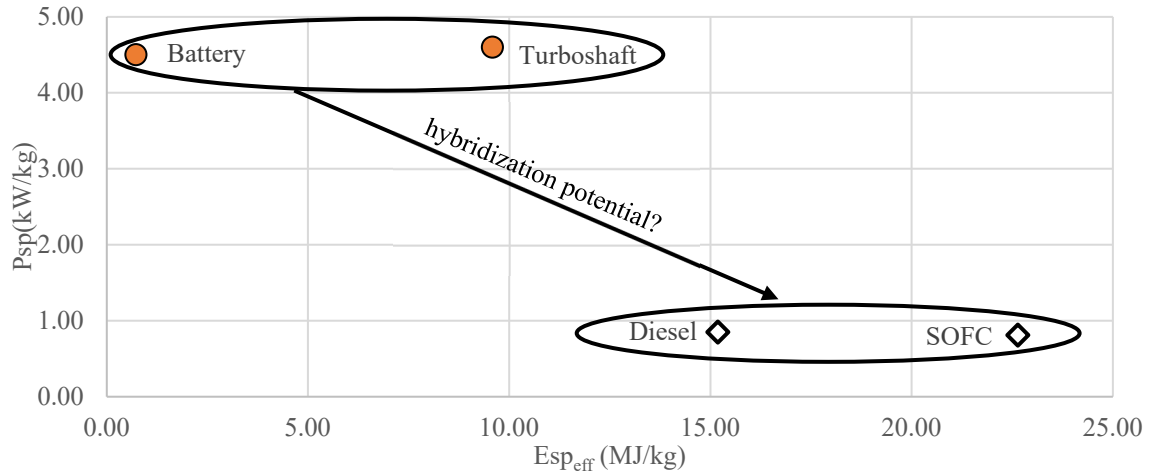


Fig. 5: Energy System Matching for Hybridization

Battery systems have lower specific energy than the other listed energy systems but may be used in combination with electric motors to offer higher specific power for transient response and short, high-power hover requirements. Listed battery specific power and effective specific energy were both lower than for turboshaft engines at the scales here. As a result, hybrid-electric turboshaft architectures do not appear to provide opportunities to reduce engine size and generate system-level weight reduction for this tailsitter configuration. Therefore, turboshaft-battery hybrids were not investigated in this paper. Note that, unlike the other systems in Table 2 increasing the battery size increases both energy and power available to the system, while for the other systems (turboshaft, diesel, SOFC), power available and energy available are decoupled, requiring separate sizing of the energy conversion system and fuel tank.

Battery properties for this paper assumed a pack-level specific energy of 300 W-hr/kg with a continuous C-rate of 15 and 20% margin for cycle life. From a sizing perspective, this means batteries were required to run the motor at maximum continuous power for four minutes. The weight for this system is identical to a system with a specific energy of 150 W-hr/kg for two minutes of hover operation (C-rate=30) when neglecting battery efficiency differences. Batteries with this combination of high specific energy (300 W-hr/kg) and high continuous C-rate operation (15C) are beyond commercially available systems at the time of writing. To further understand technology trends, a regression for pack-level specific energy vs continuous C-rate was constructed from current and emerging battery technology and pictured in Fig. 6 [30-35]. The pack overhead was an assumed 25% of cell-level specific energy. Following the trend gives an estimated pack-level specific energy of ~240 W-hr/kg for a C-rate of 15. A 4.3% improvement in specific energy per year was estimated from Ref. [36], which suggests that this combination

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of specific energy and discharge rate may be realizable sometime in the 2030 timeframe, provided technology investment remains at similar levels.

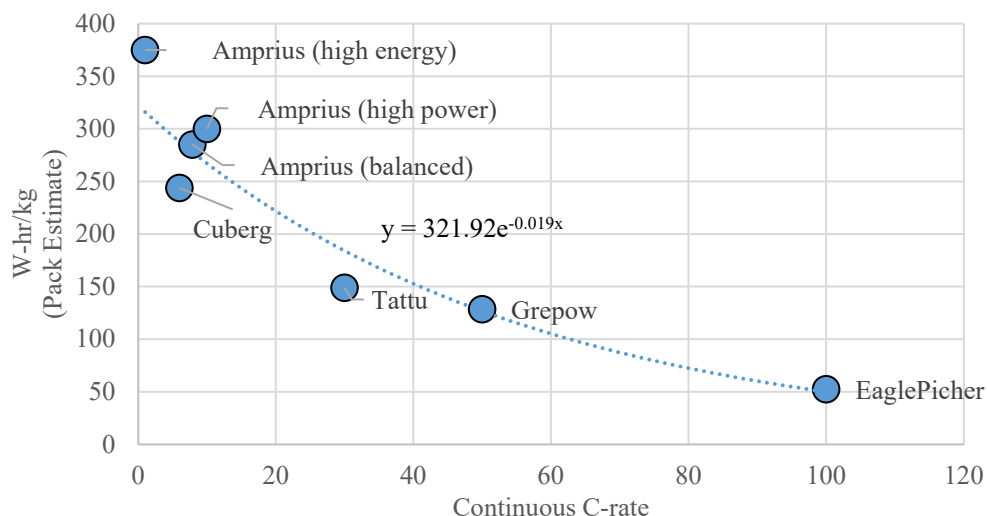


Fig. 6: Battery Technology Trend

In addition to the reference properties described in Table 2, each system possesses different off-design characteristics. Turboshfts have the lowest efficiency at part power, while diesel engines are relatively more fuel efficient at low throttle. Fuel cell efficiency is highest at part power operation. Representative efficiency vs. power for a diesel engine, a turboshaft, and a solid oxide fuel cell system can be seen in Fig. 7. Note that solid oxide fuel cell performance at part power resembles that of the more widespread proton exchange membrane (PEM) fuel cells.

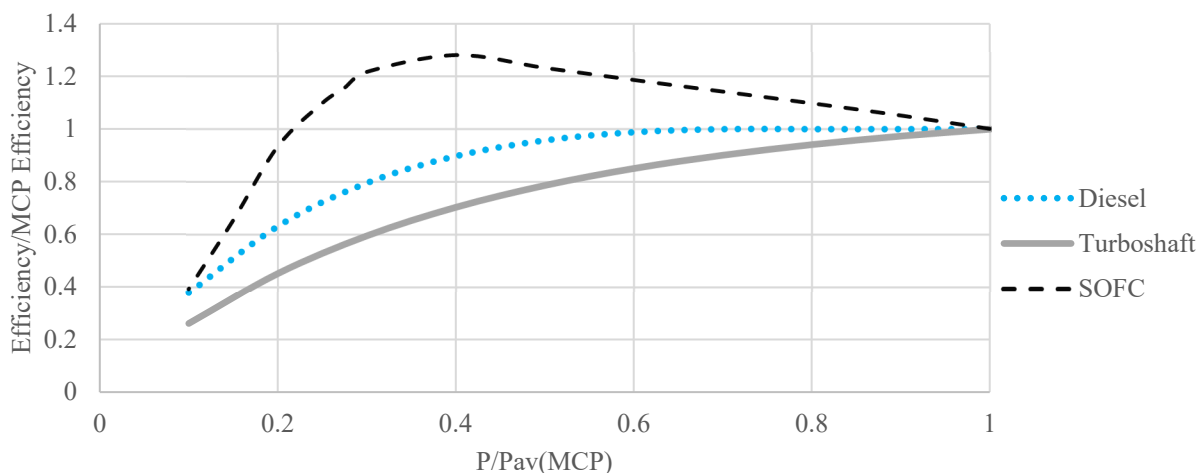


Fig. 7: Part Power Efficiency Comparison (SLS)

Aircraft that loiter at partial power for extended time periods offer potential advantages for the diesel and SOFC propulsion systems over turboshafts from a duty-cycle perspective. An additional proposed benefit of hybrid-electric architectures is the opportunity to operate as a mobile charging station for increased operational flexibility. Each of the propulsion systems analyzed in this paper offer different characteristics for these purposes, including overall conversion efficiency, required secondary equipment, fuel flexibility, as well as start time. Table 3 summarizes some of the key features of each architecture, along with required secondary equipment.

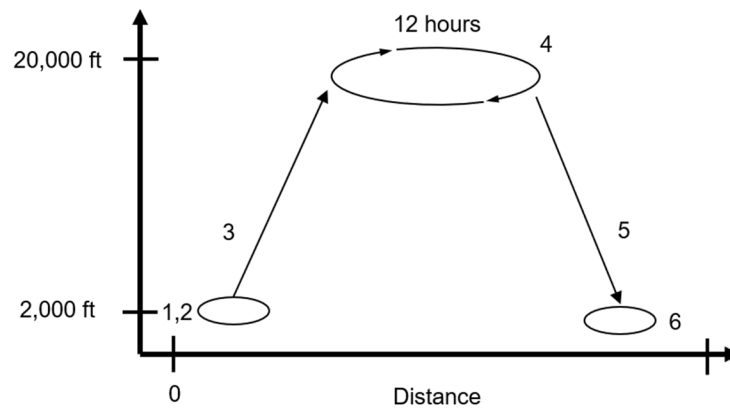
Table 3: Suitability for Emergency Ground Power

	Diesel	Diesel-Electric	Turboshaft	SOFC	
Conversion Efficiency	~37%	~37%	~22%	~60%	
Fuel	Diesel	Diesel	Jet Fuel	Jet Fuel Diesel, Other heavy fuels	H2, Propane, Methane, Natural Gas
External Equipment	Generator	None	Generator	Reformer, Desulfurizer	None
Start time	≈1 minute	≈1 minute	≈10 seconds	>≈30 minutes	

The SOFC series-hybrid architecture appears to offer compelling attributes for this purpose, including high energy conversion efficiency and fuel flexibility. SOFCs are also capable of operating on fuels such as H2, propane, methane, or natural gas with a wider variety of conventional aviation fuels when employing external reformation. Liquid heavy fuels are the most convenient from an infrastructure perspective but require external reformation for operation within the fuel cell. Fuels that are more convenient from a fuel cell reformation perspective (H2, propane, methane, natural gas), likely require additional infrastructure and may be unavailable at Army forward operating bases. Note that reciprocating and gas turbine engines may be designed and constructed to burn other fuel sources but may suffer lower efficiency or premature failure when operated on other than the intended fuel. In contrast a single SOFC stack can utilize other fuel types in the presence of a fuel reformation system. The fuel cell stack must be evenly heated for operation, requiring long startup times. The sizing mission used for this study can be seen in Table 4 and Fig. 8.

Table 4: Mission Definition

Segment	Type	Altitude Start (ft)	Time (min)	Flight Speed (kts)	Climb Speed (ft/min)
1	Idle	2,000	5	0	0
2	Hover	2,000	2	0	0
3	Climb	2,000	36	70	500
4	Loiter	20,000	720	70	0
5	Descent	20,000	36	70	-500
6	Hover	2,000	2	0	0
7	Reserve	2,000	20	70	0

**Assumptions**

- Payload: 500 lb
- Mission takeoff condition: 2,000 ft, 70 degrees F
- Vehicle accessory power: 50 hp

Fig. 8: Mission Summary

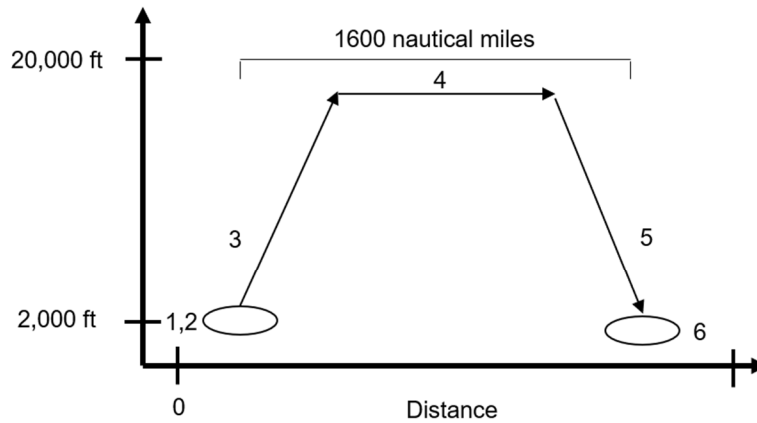
Hover was performed out of ground effect (HOGE). The diesel and turboshaft engines were sized to operate in hover at 90% max rated power (MRP). Electric motors were sized to operate in hover at 100% MCP. The fuel cell system power was determined by the highest power of Segments 3,4,5, and 7 with the battery discharging in Segment 2. The fuel cell (and diesel engine for the diesel-electric case) was unable to completely power the aircraft in hover out of ground effect at takeoff for the baseline aircraft (and all aircraft sized for this study); this design decision mitigates the weight penalty from employing these more efficient, but heavy energy conversion systems in the presence of sufficient battery and motor technology. The hybrid systems recharged the battery within

Segment 4. A 50 hp allocation for accessory power was utilized for high-power mission equipment and autonomy relevant to military applications. 20-minutes of loiter was used as a reserve segment.

This mission was constructed with the intention of offering a duty cycle where the hybrid-electric architectures are more likely to provide weight or fuel burn benefits compared to turboshaft engines due to long operation at part power in conjunction with short hover requirements. Gradient based optimization was performed across a range of loitering times to observe aircraft-level sensitivities to changes in mission requirements as well as compare the relative suitability of each propulsion architecture. In addition, vehicles were sized for fixed range, modifying Segment 4 from Table 4 to generate Table 5 and Fig. 9. For the fixed range cases, range credit was applied from Segments 3 and 5 towards Segment 4 for a total mission distance of 1,600 nautical miles at best range speed. For the loiter, cruise, and descent segments (segments 4, 5, and 7 in Table 4 and Table 5), rotor tip speed was lowered to 80% of the hover value. As in the fixed loitering time cases, optimization was performed to observe design choices that reduce DGW based on the propulsion system configuration.

Table 5: Fixed-Range Mission Definition

Segment	Type	Altitude Start (ft)	Time (min)	Flight Speed (kts)	Climb Speed (ft/min)
1	Idle	2,000	5	0	0
2	Hover	2,000	2	0	0
3	Climb	2,000	36	70	500
4	Cruise	20,000	fallout	fallout	0
5	Descent	20,000	36	70	-500
6	Hover	2,000	2	0	0
7	Reserve	2,000	20	70	0



Assumptions

- Payload: 500 lb
- Mission takeoff condition: 2,000 ft, 70 degrees F
- Vehicle accessory power: 50 hp

Fig. 9: Fixed-Range Mission Summary

IV. Results

A. Design for Long Endurance

A weights summary for each of the designs can be seen in Table 6. The diesel-electric configuration utilized a fixed 100 hp electric motor to assist in hover out of ground effect in takeoff. Differences in the wing, rotor, and fuselage weight are a result of changes in the design gross weight requirements to perform the mission. For this mission and set of propulsion system assumptions, the turboshaft-powered system was the lightest, followed by the SOFC, the diesel-electric design, and finally, the pure diesel. The SOFC-powered aircraft possessed the smallest fuel burn (~50% lower fuel burn compared to the other aircraft). The turboshaft design possessed the lowest weight and the second lowest fuel burn. The diesel-electric aircraft was lighter and consumed less fuel than the pure-diesel arrangement. The SOFC fuel tank was heavier than the other designs due to propane utilization which carries a weight penalty (see Table 2). A SOFC-powered design with the same specific power and efficiency, utilizing diesel or JP-8, would possess comparable performance to the propane-based results here; propane specific energy is higher than diesel or JP-8, but this is offset by lower tank efficiency. The primary benefit from using propane compared to traditional aviation fuels in a SOFC is to reduce balance of plant requirements; propane usage also aligns with currently fielded systems [37].

Table 6: Baseline Weight Summary (lb)

	Diesel	Diesel-Electric	Turboshaft	SOFC
DGW	7,060	6,501	4,760	5,506
Empty	5,031	4,619	2,874	4,366
-Structure	2,455	2,265	1,675	1,842
--Wing	580	532	382	446
--Rotor	520	470	321	383
--Fuselage	780	736	593	656
--Empennage	174	174	174	174
-Propulsion	1,913	1,717	661	1,924
--Engine	1,550	1,283	304	-
--Fuel Cell	-	-	-	822
--Motor	-	25	-	117
--Battery	-	54	-	385
--Drive System	459	421	310	333
-Systems	537	521	466	490
Fuel Tank	88	82	86	268
Fuel	1,521	1,370	1,376	630
Payload	500	500	500	500

Performance benefits for the diesel-electric system when compared to the pure diesel were largely dependent on duty cycle selection as well as two component level technology assumptions: the first was batteries with pack-level specific energy of 300 W-hr/kg and high discharge rates (15C). The second was high specific-torque motors at the propulsion system scales here; specific torque was based on published numbers by Joby Aviation which possess a specific torque 6X (20 ft-lb/lb) higher than the electric motors that power Tesla automobiles [9]. These two assumptions concurrently enabled the propulsion-systems tailoring performed here. Table 7 shows the power of each engine system at max rated power (MRP), max continuous power (MCP) of the motor, max rated power of the fuel cell, and the SFC at sea level standard conditions and in loiter. Recall that the diesel and diesel-

electric configurations each possessed two diesel engines; the combined power of both engines was reported for the diesel and diesel-electric aircraft.

Table 7: Engine Power and Efficiency

	Diesel	Diesel-Electric	Turboshaft	SOFC
Engine hp (MRP)	1,010	832	813	-
Motor hp (MCP)	-	100	-	755
Fuel Cell hp (MRP)	-	-	-	440
SFC _{ref} (lb/hp-hr)	0.327	0.332	0.427	0.206
SFC _{loiter,start} (lb/hp-hr)	0.426	0.410	0.536	0.201

The diesel-electric case possessed a smaller engine than the pure diesel due to the presence of the motor which offset short-term power requirements in hover. Reported engine power for the turboshaft and diesel engines is larger than for the electric motor due to the hover criteria (90%MRP) and the 2,000 ft, 70 F takeoff condition which reduces power available compared to SLS conditions. The fuel cell did not meet motor power demands in HOGE but was instead sized for climb. This required the battery to discharge during Segments 2 and 6. Recall that a constant 50 hp accessory power requirement was imposed for all mission segments, consuming a substantial fraction of power available for the fuel cell. Performance for maximum continuous and best range speed can be seen in Table 8. Best range speed, maximum range, dash speed, and dash range were estimated by taking the aircraft sized for the Table 4 mission and modifying Segment 4 from Table 5 for fallout best range or maximum speed performance.

Table 8: Baseline Tailsitter Range Comparison

	Diesel	Diesel-Electric	Turboshaft	SOFC
Best range speed (kt)	123	122	124	109
Maximum Range (nautical miles)	1,317	1,276	1,313	1,222
MCP Speed (kt)	166	160	172	148
MCP Range (nautical miles)	1,234	1,217	1,225	928

The turboshaft, diesel, and diesel-electric configurations possessed comparable range and high-speed characteristics. The turboshaft model possessed the highest MCP speed. The SOFC design possessed the worst long-range performance, with slightly lower maximum range and best range speed and substantially lower MCP speed and range. These results are due to the different propulsion systems' relative performance at part power (see Fig. 7).

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SOFC systems are most efficient at part power (such as in loiter). Because of this part-power behavior, a SOFC-powered aircraft designed for long-term operation at low power results in comparably worse performance at more power intensive applications when compared to diesel or turboshaft systems.

Next, each aircraft was optimized, with disk loading (DL), design blade loading (CW/s), wing loading (WL), tip speed, and loiter velocity selected to minimize the DGW. For the SOFC design, the fuel cell power was a design parameter as well, constrained by the largest of climb, loiter, and cruise power. The variable bounds can be seen in the Table 9. Rotor blades were constrained to aspect ratios less than 18. Optimization was performed using RCOTools and OpenMDAO as wrappers for NDARC [38, 39]. Sequential Least Squares Programming (SLSQP) was selected to perform the optimization [40]. A summary of each optimal design for the 12-hour condition can be seen in Table 10. A geometric comparison of each of the designs can be seen in Fig. 10. The diesel-electric, turboshaft and SOFC rotors were visually indistinguishable.

Table 9: Design Variables

	CW/s	DL (PSF)	WL (PSF)	Hover Tip Speed (ft/s)	Loiter Velocity (kt)
Baseline	0.12	6	15	750	70
Upper Bound	0.14	20	40	800	150
Lower Bound	0.04	4	10	400	60

Table 10: Optimization Design Summary

	Diesel	Diesel- Electric	Turboshaft	SOFC
DGW (lb)	5,560.36	4,569.04	4,462.73	4,335.25
Mission Fuel Burn (lb)	864.70	769.72	1,030.99	415.84
DL (psf)	4.00	5.67	5.85	5.31
Hover Tip Speed (ft/s)	517.79	566.72	600.65	550.18
CW/s	0.12	0.14	0.13	0.14
WL (psf)	10.03	15.50	10.00	10.00
Loiter Velocity (kt)	60.00	71.11	60.00	60.00
Peng (or Pfc) (hp)	670.25	400.09	815.50	238.88
Pmotor (hp)		143.89	-	570.10

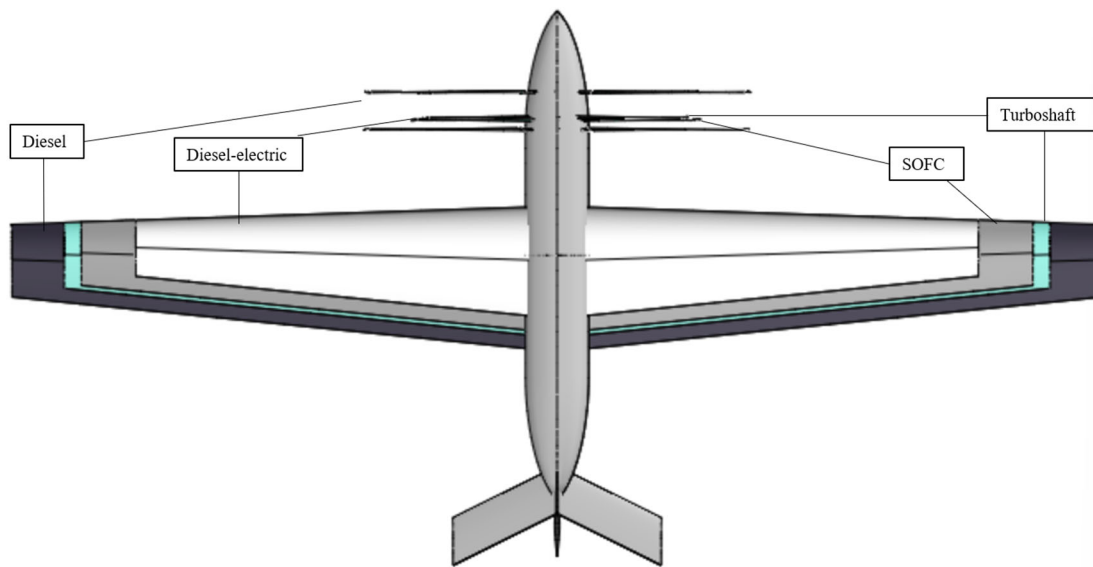


Fig. 10: Optimized Aircraft Geometry Comparison

Each aircraft possessed a combination of rotor tip speed, disk loading, and blade loading such that the rotor blade aspect ratio upper bound (18) was active. Lower disk loading results in lower power requirements in hover, reducing the engine size for the diesel and turboshaft aircraft and motor and battery size for the diesel-electric and SOFC aircraft. Higher disk loading reduces rotor size and weight and results in a better match for forward flight conditions, increasing propulsive efficiency for these aircraft. Higher wing loading reduces wing area and by extension structural weight and profile drag but increases the required pitch angle for forward flight, increasing induced drag. Lower wing loading also enables operation at lower flight speeds by providing additional stall margin. The diesel aircraft was driven to the disk loading lower boundary, which reduced the engine size for the heavy, but relatively efficient reciprocating engine (when compared to gas turbine engines). The diesel, turboshaft, and SOFC designs were each driven to the lower boundary of wing loading and loitering velocity. Key findings from these optimized designs include:

- The diesel-electric design possessed the highest optimal wing loading and loitering speed
- The diesel-electric design possessed a larger electric motor to assist in hover out of ground effect compared to the baseline version
- The turboshaft-powered design possessed the highest disk loading and fuel burn.

- The SOFC configuration maintained approximately 50% lower fuel burn compared to the other designs (similar to the baseline cases).
- The diesel and diesel-electric designs showed the greatest reduction in gross weight compared to the equivalent baseline (see Table 6).
- Weight reduction in the diesel-electric and SOFC designs was primarily due to a reduction in climb power, enabling smaller, lighter energy conversion systems.

Compared to the turboshaft engine design, the diesel-electric and SOFC designs reduced fuel burn by about 30% and 60%, respectively. The diesel-electric and SOFC architectures partially decoupled power requirements between hover and forward flight; this creates freedom in the design space to accept a less efficient hovering condition for these short time scales and to take advantage of more efficient (but heavier) energy conversion systems for forward flight by leveraging the electric system for short duration, high power conditions. Higher disk loading for the turboshaft increased the engine size, resulting in improvements in power/weight as well as nominal SFC due to favorable scaling (see Eq. (1) and (2)). Optimized DGW of each aircraft configuration vs. changing loitering time can be seen in Fig. 11, with mission fuel burn plotted in Fig. 12.

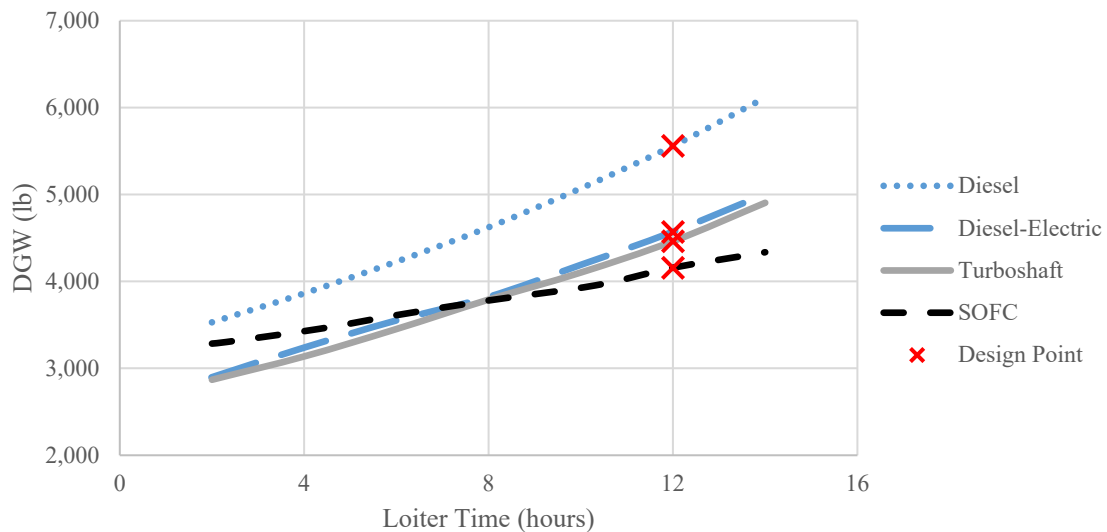


Fig. 11: Loitering Time vs. DGW (Optimized Configuration)

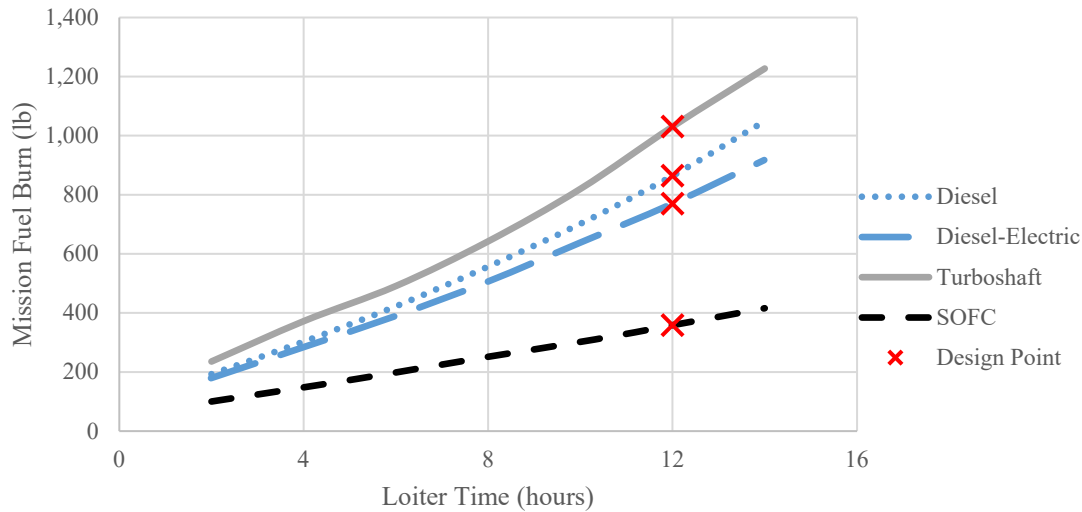


Fig. 12: Fuel Burn vs. Loiter Time (Optimized Aircraft)

For this high-level optimization study, the turboshaft was the lightest aircraft below ~8 hours of loiter time and exhibited comparable design gross weight to the diesel-electric aircraft across a range of loitering times. The SOFC series hybrid configuration consistently demonstrated the lowest fuel burn. Below ~8 hours of loitering time, the diesel-electric aircraft was lighter than the SOFC design. The diesel-electric aircraft was lighter than the pure diesel aircraft for all cases due to the short hover time requirement combined with advanced battery and electric motor assumptions. For increasing flight time, fuel becomes a larger proportion of the DGW for a given energy system; as a result, the relative potential performance benefits for hybridization increase both from DGW as well as fuel burn perspectives. A comparison of optimized diesel-electric designs with varying motor and battery technology can be seen in Fig. 13.

For a range of battery and motor technology levels, parallel hybrid architectures that combine an electric motor with a mechanical drivetrain appears to allow for improvements when comparing the diesel-electric configuration to a pure diesel configuration. Improved battery technology or reduced hover time seem to be required to achieve lower DGW than the turboshaft-powered designs. In addition, for a given battery technology level, there appears to be a motor specific torque threshold such that further improvements to torque/weight produce

diminishing weight reduction returns. Note that all battery systems were sized for four minutes of hover operation (or a 15C battery).

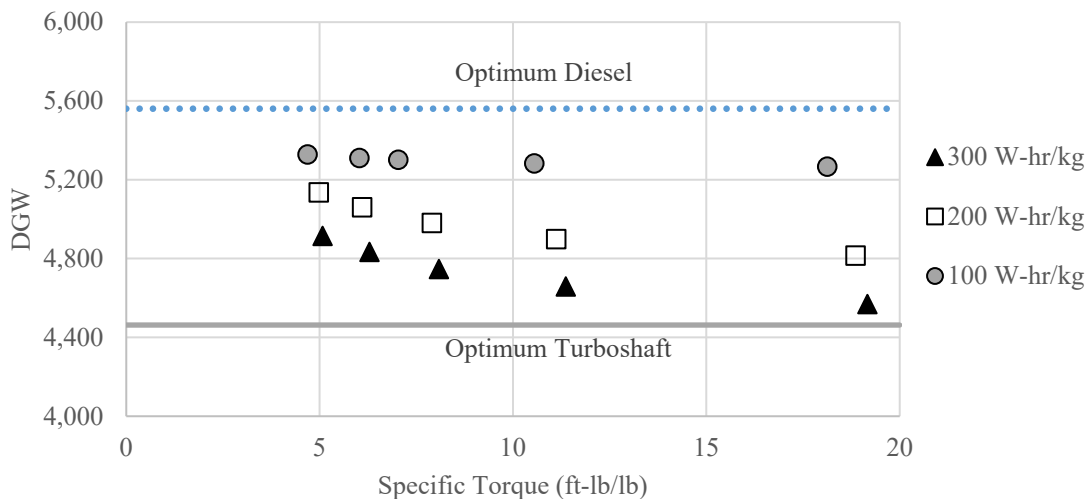


Fig. 13: Diesel-Electric Technology Sensitivity (12 hour loiter)

Varying hover time requirements allow for the DGW results here to be roughly comparable across different battery types. For example, a battery sized for two minutes instead of four minutes of hover increases the C-rate to 30; a 150 W-hr/kg battery at 30C is then equivalent to a 300 W-hr/kg battery at 15C (neglecting C-rate impacts on efficiency). Pack-level arrangement of cells (series vs parallel) also presents some opportunities for tailoring performance based on design needs; these trades were not investigated as part of this paper. In general, minimization of hover time requirements is critical to obtaining design benefits for these hybrid-electric configurations. Recall that selection of this particular mission set and use case (short hover times coupled with extended low-speed operation) was intended to provide cases where hybrid-electric systems may be competitive with turboshaft-powered systems to further understanding of the potential design space. The following section explores fixed-range applications and sensitivities for these different propulsion architectures.

B. Design for Fixed Range

The optimization problem from Table 9 and Table 10 was repeated, using the mission described in Table 5 and Fig. 9. Cruise velocity for Segment 4 was added as an additional design variable and each aircraft configuration was optimized, minimizing DGW. Results for a fixed 1,600 nautical mile range mission can be seen in Table 11, with a

visual comparison of the different aircraft shown in Fig. 14. For the fixed range cases, one can observe the following differences in the optimal designs:

- As in the fixed loiter time cases, the diesel aircraft possessed the lowest disk loading, reducing relative engine system power and weight.
- The turboshaft and SOFC-powered aircraft possessed considerably higher disk loading than for the fixed loiter time mission.
- The turboshaft design possessed the highest DL, fuel burn, and cruise speed.
- The SOFC design possessed the lowest DGW, fuel burn, and cruise speed.

Table 11: Optimization Comparison (Fixed 1,600 Nautical Mile Range)

Optimization Comparison	Diesel	Diesel-Electric	Turboshaft	SOFC
DGW (lb)	5,732.98	5,061.15	4,669.86	4,176.54
Mission Fuel Burn (lb)	1,065.03	988.57	1,316.14	504.92
DL (psf)	4.00	5.96	8.60	7.73
Rotor Tip Speed (ft/s)	522.15	582.84	741.96	661.88
CW/s	0.12	0.14	0.12	0.14
WL (psf)	25.30	19.84	36.68	19.57
Cruise Velocity (kt)	142.18	121.91	155.57	109.96
Loiter Velocity (kt)	86.78	88.92	111.00	78.59
Peng (or Pfc) (hp)	688.61	464.68	824.77	241.02
Pmotor (hp)	-	152.57	-	681.59

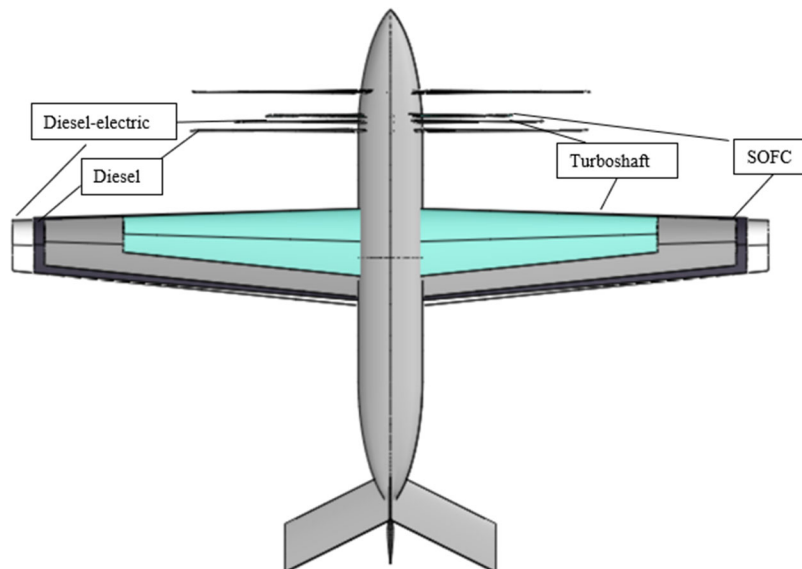


Fig. 14: Optimized Aircraft Geometry Comparison (Fixed Range)

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Rotor and aerodynamic performance models used for this study predicted higher propulsive efficiency and lower rotor hub drag when employing a higher disk loading at higher airspeeds. The rotor blade aspect ratio constraint was active for all optimized designs and would increase if relaxed. Note that, with the fixed empennage geometry selected for this study, control characteristics for the different aircraft would vary substantially. These design choices were selected to enable consistent comparison of these propulsion options; configurational changes may be warranted for these aircraft prior to more detailed analysis or development.

For the hybrid-electric cases, the optimizer reduced DGW by downsizing the primary energy conversion system and utilizing a motor and battery for short hover segments. This design choice resulted in lower fuel burn than the pure diesel and turboshaft-powered aircraft due to lower gross weight (compared to the diesel) or increased energy conversion system efficiency (compared to the turboshaft). However, reducing the size of the main energy converter resulted in lower power available in cruise, yielding lower optimal forward flight speed compared to the single engine system aircraft. Differences in the optimal disk loading between the diesel-electric and SOFC designs stemmed from their differing hybrid architectures, i.e. series (SOFC) vs parallel (diesel-electric). In the diesel-electric design, increasing engine size reduces battery size, while for the SOFC design, the battery was dictated solely by hover power. Recall that the payload for these aircraft was 500 lb; the turboshaft aircraft would likely become more competitive compared to the other energy system choices with increasing payload due to beneficial scaling effects with increasing power and a relative reduction in fuel fraction for the available useful load.

Except for the pure diesel architecture, disk loading and rotor tip speed were higher for the fixed range optimization cases than the fixed loitering time cases. Wing loading for the fixed range optima also increased. Cruise speed and loiter speed were each considerably greater for the fixed range cases than the fixed loiter time cases. Rotor and wing design choices were reflective of the presence of flight speed as a driver when range is included as a requirement. In contrast, for the fixed loitering cases, minimization of power requirements was the primary factor. A robust optimization framework was required to explore the design space and rigorously evaluate these tradeoffs for the hybrid aircraft configurations due to the need to balance multiple energy systems. A comparison of aircraft design gross weight and mission fuel burn for design ranges between 200-1600 nautical mile can be seen in Fig. 15 and Fig. 16.

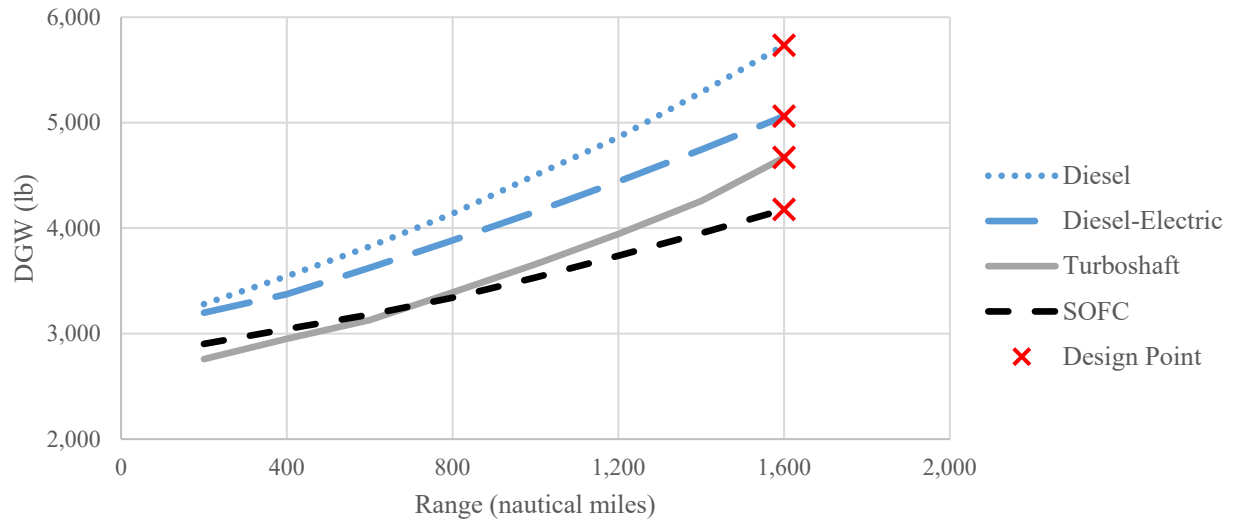


Fig. 15: Range vs. DGW (Optimized Configuration)

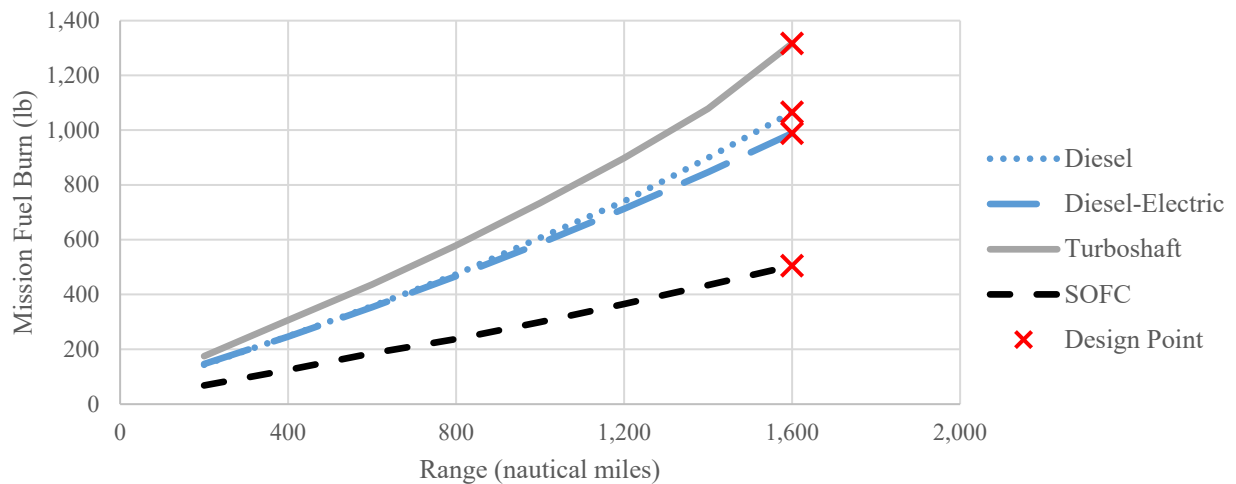


Fig. 16: Fuel Burn vs Range (Optimized Configuration)

Below ~700 nautical miles of range, the turboshaft design was the lightest aircraft, while the SOFC design was lighter at longer ranges. Consistent with the fixed loitering time results, the turboshaft designs possessed the highest fuel burn, followed by the diesel, the diesel-electric, and finally the solid oxide fuel cell aircraft. With increasing range, weight and fuel burn gaps between the diesel and diesel-electric designs grew. In addition, increasing range resulted in greater DGW and fuel burn performance benefits for the hybrid-electric propulsion architectures compared to the turboshaft and diesel cases. Note that the optimized diesel-electric aircraft possessed lower cruise speeds than the diesel designs for the results in Fig. 15 and Fig. 16. An important caveat is that, while the

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comparisons across the different energy systems were consistent, the optimizer performed large scale changes to the wing and rotor spanning large differences in both weight and geometry. While this is appropriate for conceptual-level studies that investigate trends and technology sensitivities, higher fidelity analysis and further component-level performance substantiation is recommended for any potential follow-on work.

V. Conclusions

Tailsitter designs using four separate propulsion system architectures were compared and optimized for both fixed loitering time and fixed range use cases. The hybrid-electric propulsion systems presented opportunities for design flexibility by sizing more efficient, but heavier energy conversion systems for climb and forward flight while utilizing batteries and electric motors for short-duration hover. Solid-oxide fuel cell systems appear to possess compelling characteristics for long endurance performance with comparable system-level weight to turboshaft-powered aircraft while burning roughly half the fuel. Projected performance for the SOFC systems was dependent on the maturation of improved power density solid oxide fuel cell stacks, application of a combined cycle approach in an aviation platform, or both. The other hybrid configuration (diesel-electric) showed improved performance over the pure diesel design but required the introduction of advanced battery technology to become competitive with a turboshaft architecture from a weight perspective. Both optimized hybrid configurations possessed considerably lower fuel burn for long range and long endurance requirements when compared to pure turboshaft designs and considerably lower aircraft weight compared to pure diesel designs, but with lower forward flight speed. Maintaining short hover time was essential to prevent battery weight growth from outpacing potential benefits for these configurations. These results suggest that, with sufficient battery and electric motor technology, hybrid propulsion architectures can offer the potential for fuel efficient, long range, long endurance, VTOL capable aircraft in instances where low forward flight speed and short hover times are acceptable.

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